TEXORA · LECTURE SHEET

Application Areas of Technical Textiles

The syllabus names five fields — smart, medical, geo, military and protective textiles — taken in turn here, with the materials, production methods and governing properties for each, before closing with the wider twelve-area map.

TE-4206 · TECHNICAL TEXTILE AND NANO-TECHNOLOGY · TOPIC 4 · LEVEL 4 TERM 2

1. How to read an application area

For each area the engineering questions are the same: **what material, made by what method, delivering which property, judged by which test?** The five fields below are treated in that order. They were named in the syllabus as examples (“etc.”) — the remaining areas follow the same pattern and are summarised in Section 7.

2. Smart textiles

Definition. Textiles that sense, react to, or adapt to environmental conditions or stimuli — heat, light, moisture, pressure, electrical signals. A useful ladder of sophistication:

- **Passive smart** — senses only: a fabric with embedded optical fibres or conductive yarns that measure strain or temperature.
- **Active smart** — senses and reacts: a garment that warms when cold (resistive heating elements), a wound dressing that releases a drug on infection signals.
- **Very smart** — senses, reacts and adapts: systems that learn the wearer's pattern and adjust. Mostly research-stage.

Materials and production. Conductive yarns (silver-coated polyamide, stainless-steel blends, carbon), conductive polymers and printed conductive inks; sensors knitted or woven in during fabric formation, or laminated/printed afterwards; energy from flexible batteries or energy harvesting. The textile route matters: integrating a sensor *during* knitting or weaving gives durability that glued-on electronics rarely match.

Governing properties and honest cautions. Signal stability after washing and flexing is the weak point of the whole field — a sensor that drifts after five washes is a demonstration, not a product. Power supply, connector reliability and wash-care labelling decide commercial success more than the sensing principle. Claims in this field should always be asked: *after how many washes, measured how?*

3. Medical textiles (Medtech)

Medical textiles divide by where they serve:

FAMILY	EXAMPLES	DECIDING PROPERTIES
Implantable	Sutures, vascular grafts, hernia meshes, artificial ligaments	Biocompatibility, strength retention in the body, sterility, controlled degradation (absorbable) or permanence
Non-implantable	Wound dressings, bandages, plasters, pressure garments	Absorbency, moisture-vapour transmission, non-adherence, barrier to bacteria
Healthcare / hygiene	Surgical gowns and drapes, masks, caps, wipes, bedding	Barrier performance, lint control, disposability economics, sterilisation tolerance
Extracorporeal devices	Artificial kidney/liver membranes and supports	Controlled permeability, blood compatibility

Table 1 — The four Medtech families. Materials: cotton and viscose for absorbency; polyester, polypropylene and PTFE for implants and barriers; alginate and chitosan fibres for advanced dressings; production by weaving, knitting, braiding and nonwoven routes, in clean conditions for implantables.

Worked principle — why absorbency is a number, not an adjective. A dressing is specified by absorbency (grams of fluid per gram of dressing, or per 100 cm^2) and by moisture-vapour transmission rate (MVTR, $g/m^2/24$ h): too low an MVTR macerates the wound, too high dries it. Selection in Medtech is always this pattern — a window of acceptable values, tested by named methods, inside a regulatory framework that treats the product as a medical device.

4. Geotextiles (Geotech)

Definition. Permeable textile materials used in contact with soil, rock or water in civil engineering. Four classical functions organise the whole subject:

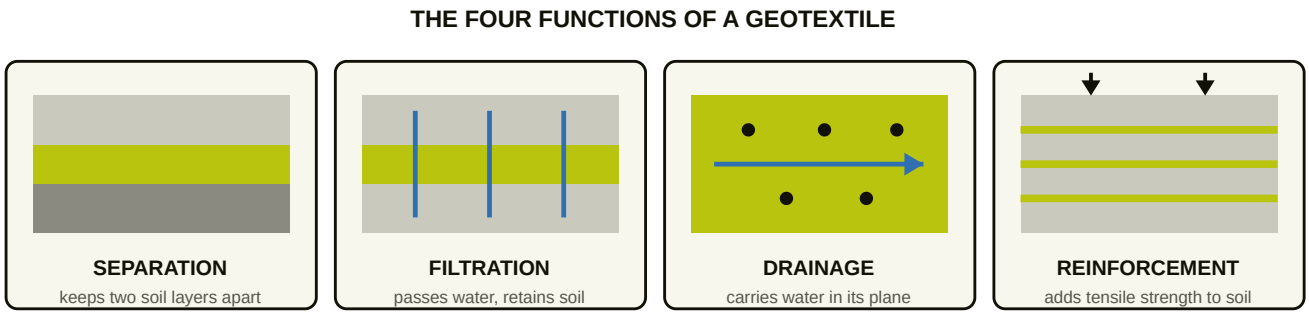


Figure 1 — Separation, filtration, drainage and reinforcement: the four functions around which geotextile specifications are written.

Materials and production. Polypropylene and polyester dominate (chemical resistance in soil, cost); woven slit-film and multifilament fabrics for strength-led duties, needle-punched nonwovens for filtration and protection, knitted and grid structures (geogrids) for reinforcement. **Governing properties:** tensile strength and elongation, pore size (to retain soil yet pass water), permeability, puncture resistance, and durability in soil over design lives measured in decades — which is why Topic 3's creep and chemical screening matter here most of all.

5. Military textiles

Military textiles are conventional and technical products pushed to simultaneous extremes: camouflage (visual, and near-infrared reflectance matched to terrain so the wearer does not glow under night-vision), ballistic and fragment protection, flame resistance, chemical/biological protective ensembles, parachutes and aerial delivery, tents and shelters for all climates. **Materials:** para-aramid for ballistics, high-tenacity polyamide for parachutes, flame-resistant fibres and finishes for combat clothing, activated-carbon fabrics for chemical protection. **Production:** weaving and finishing to defence specifications with lot-by-lot testing. **The governing idea:** multi-threat design — one garment system must manage ballistics, weather, flame and signature together, and weight is a combat property in itself: every gram is carried by a person.

6. Protective textiles (Protech)

Protection is organised by hazard:

HAZARD	TEXTILE ANSWER	GOVERNING PROPERTY / TEST IDEA

Heat and flame	Meta-aramid, FR-treated cotton, modacrylic blends; aluminised suits for radiant heat	Flame spread, thermal protective performance, no melting/dripping
Ballistic / stab	Para-aramid or HMPE woven/UD layers in vests and helmets	Energy absorption across many layers; back-face deformation limits
Cuts	HMPE, glass-core and steel-core yarns in gloves and sleeves	Cut resistance under standard blade tests
Chemicals	Coated and laminated barrier fabrics, butyl and multilayer ensembles	Permeation breakthrough time for named chemicals
Electrical / arc	FR arc-rated fabrics; conductive antistatic fabrics	Arc rating (cal/cm²), surface resistivity
Falls	Polyamide/polyester webbing in harnesses and lanyards	Breaking force with energy-absorbing deployment
Visibility	Fluorescent backgrounds with retroreflective tapes	Photometric performance day (colour) and night (reflectivity)

Table 2 — Protech organised by hazard. In protective clothing the standard is the syllabus: products are certified against named protective standards, and an uncertified claim is not a protective product.

Layering principle. Serious protection is a system: an outer shell (abrasion, flame), a barrier layer (liquid/vapour), a thermal liner, and a comfort layer. Ballistic vests make the same point mechanically — no single layer stops the projectile; twenty or thirty layers, each catching and spreading energy, do it together.

7. The rest of the map, briefly

Agrotech — shade and hail nets, crop covers: UV-stable PP/PE, light transmission by design. **Buildtech** — membrane architecture (PTFE/glass or PVC/polyester coated fabrics), scaffolding nets. **Clothtech** — interlinings, threads, linings: the invisible technical content of ordinary garments. **Hometech** — mattress and furniture components, flame-retardant contract furnishing. **Indutech** — filtration above all: bag filters of PPS/PTFE for hot gas, liquid filter cloths, conveyor and drive belts. **Mobiltech** — seat belts and airbags (high-tenacity PET and PA), tyre cord, automotive interiors and insulation. **Oekotech** — oil booms and absorbents, erosion-control mats, landfill drainage composites. **Packtech** — FIBC bulk bags of woven PP, tested by rated load and safety factor. **Sporttech** — sails, parachutes (leisure), tents, artificial turf, climbing ropes with dynamic energy absorption.

Glossary

MVTR — moisture vapour transmission rate; the breathing rate of a fabric or dressing. **Biocompatibility** — the property of not harming living tissue in the intended contact. **Geogrid** — an open grid structure used to reinforce soil. **Back-face deformation** — the dent a stopped projectile makes in a backing material; limited because the body is behind it. **Permeation breakthrough time** — time for a chemical to pass a barrier fabric at a defined detection rate.

Key points

• Read every area the same way: material → method of production → governing property → test.
• Smart textiles climb a ladder: passive (sense) → active (sense + react) → very smart (adapt); wash durability is the field's weak point.
• Medtech is organised by place of service — implantable, non-implantable, healthcare/hygiene, extracorporeal — inside device regulation.
• Geotextiles do four jobs: separation, filtration, drainage, reinforcement — and must keep doing them for decades in soil.
• Military textiles are multi-threat and weight-critical; Protech is hazard-by-hazard and standard-by-standard, with layering as the working principle.

Practice

1. A needle-punched nonwoven is placed between a road's crushed-stone base and the soft clay below, and must let groundwater pass upward without letting clay migrate. Name its two governing functions and the two fabric properties that must be specified for them.
2. Explain why a ballistic vest is built from many layers of fabric rather than one layer of the same total mass.
3. A start-up claims its knitted shirt is a “very smart textile” because it contains a temperature sensor. Evaluate the claim against the ladder in Section 2.

Answer key

1. **Separation** (keeping stone and clay apart — specified by tensile/puncture strength so it survives installation and traffic loads) and **filtration** (passing water while retaining soil — specified by pore size/apparent opening size together with permeability). The two property pairs are: strength properties for separation; pore size + permeability for filtration.
2. Each layer absorbs part of the projectile's energy by fibre stretching, yarn pull-out and friction, and spreads the remaining load over a wider area of the next layer. A single thick layer of equal mass behaves stiffly, engages fewer yarns, and fails by localised yarn rupture; the layered system

multiplies the number of yarns and interfaces doing work, and also limits back-face deformation progressively.

3. The claim overstates the product. A temperature sensor alone makes the shirt at most **passive smart** (it senses). “Very smart” requires sensing, reacting *and adapting* its behaviour — e.g. adjusting heating or ventilation in response to learned patterns. The honest label is a sensor-equipped (passive smart) textile, and the next questions are wash durability and signal stability.
